

Yanghao Chen

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Education

Tongji University

Bachelor of Engineering in Engineering Mechanics

GPA: 87/100

Shanghai, China

Sep. 2023 – Present; expected Jun. 2027

Johns Hopkins University

Visiting Undergraduate Research Intern, Department of Civil and Systems Engineering

Baltimore, MD, USA

Jan. 2026 – Oct. 2026

Research Interests: Solid Mechanics, Scientific Machine Learning, Neural Operators.

Research Projects

GPU-Accelerated Hybrid FEM–Neural Operator Coupling Framework

Jan. 2026 – Present

Lead Undergraduate Researcher; Advisor: [Prof. Somdatta Goswami](#)

Johns Hopkins University

- Generated displacement and stress data using GPU-accelerated JAX-FEM simulations under prescribed interface displacements for neural-operator training.
- Trained Transolver in JAX to learn mappings from interface displacement fields to stress fields within the local subdomain.
- Developed a hybrid FEM–neural operator solver based on domain decomposition, with iterative exchange of interface displacements and reaction forces computed from predicted stresses.

PINNs for Shock Capturing in the Burgers Equation

Nov. 2025 – Mar. 2026

Undergraduate Researcher; Advisor: [Prof. Xianyang \(Tom\) Chen](#)

Tongji University

- Trained a PINN in PyTorch to predict shock propagation in the one-dimensional inviscid Burgers equation, using PDE-residual, initial-condition, and boundary-condition losses.
- Added constant artificial viscosity to the PDE residual, using automatic differentiation to compute the required derivatives.
- Compared predicted shock profiles with the analytical inviscid solution, examining transition width and local deviations near the discontinuity.

ConvLSTM Modeling of Chemo-Mechanical Fields in Battery Materials

Jun. 2025 – Oct. 2025

Undergraduate Researcher; Advisor: [Prof. Ying Zhao](#)

Tongji University

- Automated polycrystalline NMC microstructure generation and MATLAB–COMSOL simulations to produce time-aligned images of lithium concentration and von Mises stress.
- Trained separate conditional ConvLSTM models in PyTorch to predict future sequences of concentration and stress fields from past field images, grain-orientation maps, and C-rate inputs.
- Used MSE and SSIM losses with scheduled sampling and patch-based training for multi-step autoregressive prediction.

Domain-Specific LLM Fine-Tuning for Mechanics of Materials

Sep. 2024 – Apr. 2025

Undergraduate Researcher; Advisor: [Prof. Ying Zhao](#)

Tongji University

- Curated instruction–response datasets from textbooks and technical literature on stress analysis, constitutive laws, and failure theories.
- Fine-tuned Qwen2.5-7B with LoRA on Google Colab for mechanics of materials Q&A and concept explanations.
- Released two instruction datasets and two Qwen2.5-based model checkpoints on [Hugging Face](#).

Technical Skills

Scientific Machine Learning

PyTorch, JAX, Neural Operators (Transolver, DeepONet), PINNs, ConvLSTM

Computational Mechanics

Finite Element Methods, FEniCSx, Gmsh, COMSOL Multiphysics, ABAQUS

Scientific Computing & Tools

Python, MATLAB, Linux, Git/GitHub, LiveLink for MATLAB

LLM Fine-Tuning

Qwen2.5, LoRA/PEFT, Instruction Dataset Curation

Languages

Chinese (native), Cantonese, English